UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	OCTOBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (October 2026).
   Computed answers show working; every answer is produced by the deterministic MAT133 generator and verified by code where possible. No model answer is presented as official.

2. Every marking point comes from the generator's worked solution; M1/A1 codes are aligned from the final answer backwards.

3. Five questions; total = 70 marks.

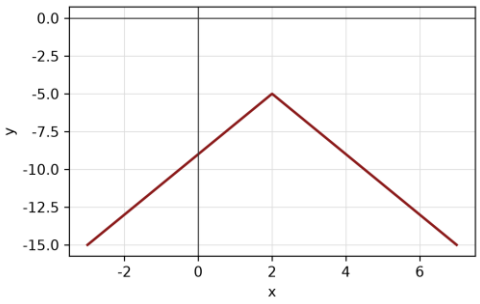
4. Computed: 15 part(s). Placeholders: 0.
   Questions and answers are for local study use only.

5. Generated deterministically from the 41 MAT133 past papers; seed 3.

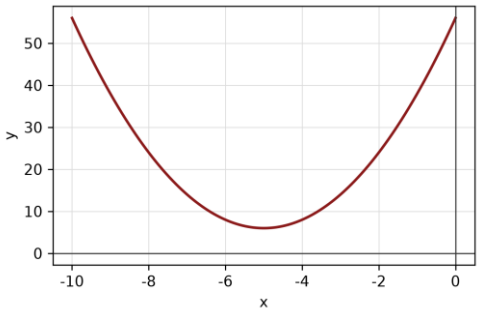
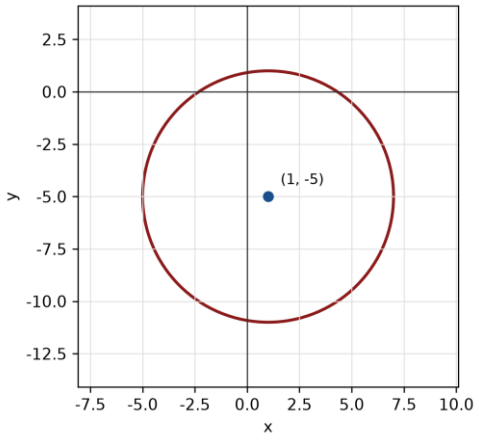
FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

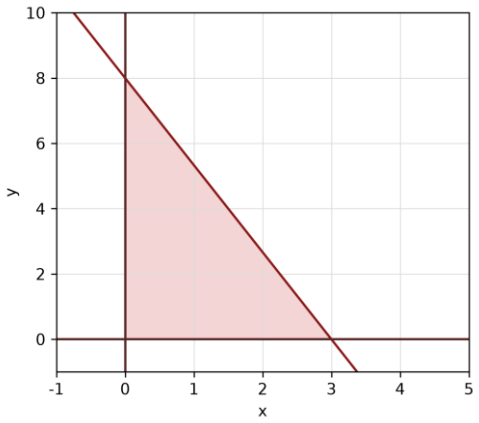
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NO	SOLUTION		MARK
1 a)	$-3x-4<-4$ $-3x<0$ $(0,\infty)$ 	M1 A1	3 m
1 b)	$\lvert 4x-5\rvert-4<0$ $-4<4x-5<4$ $\left(\frac{1}{4},\frac{9}{4}\right)$ 	M1 A1 A1	5 m
1 c)	$y=-2\lvert x-2\rvert-5$ vertex $(2,-5)$ no real intercepts 	B1 A1 A1	5 m

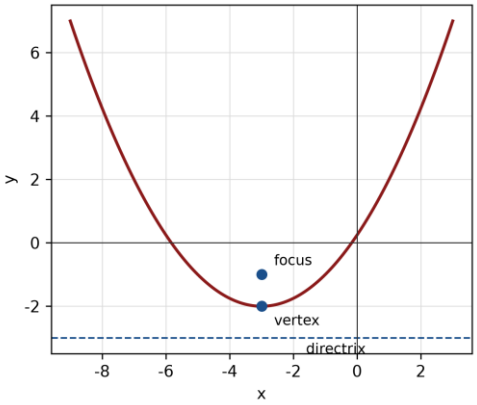
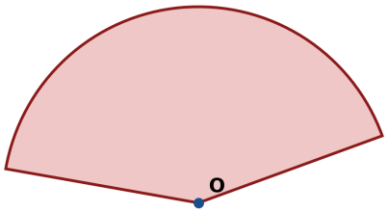
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NO	SOLUTION		MARK
2 a)	$y = 2(x+5)^2 + 6$ vertex $(-5, 6)$ $y(0) = 56$ 	B1 B1 A1 A1	5 m
2 b)	$f(x) = x^5 - 3x^3 - 6x$ $f(-x) = -x^5 + 3x^3 + 6x$ $-f(x) = -x^5 + 3x^3 + 6x$ $f(-x) = -f(x)$	M1 A1 A1	4 m
2 c)	$x^2 - 2x + y^2 + 10y = 10$ $(x-1)^2 + (y+5)^2 - 26 = 10$ $(x-1)^2 + (y+5)^2 = 36$ 	M1 M1 A1 A1	5 m

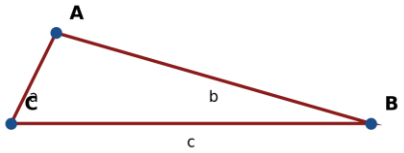
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NO	SOLUTION		MARK
3 a)	$\frac{-3+3i}{1-i}$	M1	5 m
	$= \frac{(-3+3i)(1+i)}{(1-i)(1+i)}$	M1	
	$= \frac{-6}{2}$	A1	
	-3	A1	
3 b)	$\lvert z\rvert = \sqrt{8^2+15^2} = 17$	M1 A1 M1	5 m
	$\tan\theta = \frac{15}{8}, \quad \theta = 61.9^\circ$	A1	
3 c)	$-3x^3 - 5x^2 + x + 3 = (x+4)(-3x^2+7x-27) + 111$	M1 M1 A1	5 m
	$q(x) = -3x^2 + 7x - 27, \quad r = 111$	A1	
4 a)	$\frac{x}{3} + \frac{y}{8} = 1$	M1 A1	5 m
	$x \geq 0, \quad y \geq 0, \quad 8x + 3y \leq 24$	A1	
			
4 b)	$-6x - 12 = x^2 - 4$	M1	5 m
	$x^2 + 6x + 8 = 0$	A1	
	$(x+2)(x+4) = 0$		
	$x = -4$ or $x = -2$		
	$(-4, 12), \quad (-2, 0)$	A1	

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NO	SOLUTION		MARK
4 c)	$y = \frac{x^2}{4} + \frac{3x}{2} + \frac{1}{4}$ $y = \frac{(x+3)^2}{4} - 2$ $(x+3)^2 = 4y + 8$ 	M1 A1 A1 A1	5 m
5 a)	$\cos A = -\frac{12}{13}, \quad \sin B = -\frac{5}{13}$ $\sin 2A = 2\sin A\cos A = -\frac{120}{169}$ $\cos(A+B) = \cos A\cos B - \sin A\sin B = -\frac{119}{169}$	M1 M1 A1 A1	5 m
5 b)	$\theta = 150^\circ \times \frac{\pi}{180} = \frac{5\pi}{6}$ $s = r\theta = 13\left(\frac{5\pi}{6}\right) = \frac{65\pi}{6}$ 	M1 M1 A1	4 m

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NO	SOLUTION	MARK
5 c)	$\frac{a}{\sin A} = \frac{b}{\sin B}$ **M1** $\frac{13}{\sin 45^\circ} = \frac{b}{\sin 60^\circ}$ **M1** $b = \frac{13\sin 60^\circ}{\sin 45^\circ} = \frac{13\sqrt{6}}{2}$ **A1** 	4 m

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